Deep Neural Networks — Quiz Answer Key

WILP M.Tech (AIML), BITS Pilani — with brief solving explanations

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Q1. Perceptron mistake-bound theorem $O((R/\gamma)^2)$: if every input vector AND the margin are both scaled by 10×, the mistake bound...

Answer: d. Stays unchanged, since R/γ is scale-invariant

Explanation: Scaling x by 10 scales R by 10; scaling margin γ by 10 too. The ratio R/γ is unchanged, so the bound $(R/\gamma)^2$ is unchanged.

Q2. Why does convexity matter for cross-entropy+sigmoid vs squared-error+sigmoid in single-neuron binary classification?

Answer: c. With squared-error+sigmoid, saturated sigmoid regions can cause vanishing gradients and slow/suboptimal convergence

Explanation: Squared-error loss composed with sigmoid is non-convex; in saturated regions (sigmoid output near 0 or 1) the derivative sigma'(z) ≈ 0, killing the gradient. Cross-entropy's gradient cancels this saturation term, giving a well-behaved (convex) landscape.

Q3. Why does mini-batch SGD noise often help generalization vs full-batch GD?

Answer: b. Stochastic noise acts as implicit regularization, avoiding overly sharp minima

Explanation: The gradient noise in mini-batch updates prevents the optimizer from settling into narrow/sharp minima that overfit training data, biasing it toward flatter minima that generalize better.

Q4. Linear regression trained without scaling 'age' (0-80) and 'income' (100,000-5,000,000). Main problem?

Answer: a. Gradient descent converges very slowly or oscillates due to an elongated error surface

Explanation: Very different feature scales create an elongated, ill-conditioned loss contour (like a narrow valley). Gradient descent then zigzags across the narrow dimension and converges slowly.

Q5. Logits z=[2.0, 1.0, 0.1], true class 0 (y=[1,0,0]). Find $\partial L/\partial z1 = p1 - y1$.

Answer: c. 0.242

Explanation: Softmax: p1 = e^1.0 / (e^2.0 + e^1.0 + e^0.1) = 2.718 / (7.389+2.718+1.105) = 2.718/11.212 ≈ 0.2424. Gradient = p1 - y1 = 0.242 - 0 = 0.242.

Q6. Wake-word detection (present/absent) from labeled audio clips — problem type?

Answer: b. Supervised classification

Explanation: Labeled examples (contains wake word / doesn't) with a discrete decision → supervised binary classification.

Q7. K=3 mutually exclusive classes modeled via 3 independent sigmoids (no softmax); P(A)=0.9, P(B)=0.85, P(C)=0.7. What's true?

Answer: a. Scores don't sum to 1 — independent sigmoids model each class as a separate binary event, not a joint categorical distribution

Explanation: Each sigmoid output is P(class_i present) treated as an independent yes/no decision; there's no normalization constraint forcing them to sum to 1, unlike softmax which enforces a joint probability distribution over mutually exclusive classes.

Q8. Wake-word data: ~1 positive per 10,000 negatives. Plain accuracy-maximizing training will most likely...

Answer: d. Collapse to a degenerate always-negative classifier, since that already minimizes error while ignoring the rare positive class

Explanation: With extreme class imbalance, predicting 'negative' always gives ~99.99% accuracy. Since accuracy is the objective, gradient descent has little incentive to learn the rare positive pattern.

Q9. Company wants to auto-identify an object in photos, learning useful features automatically from labeled images — best approach?

Answer: a. Deep Learning

Explanation: Deep learning (e.g., CNNs) automatically learns hierarchical feature representations from raw labeled image data, unlike manual rule-based or purely unsupervised approaches.

Q10. Perceptron: w1=1, w2=1, b=-1.5, step activation (output 1 if z>=0). Which input gives output 1?

Answer: c. (1,1) -> z = 0.5

Explanation: z = w1*x1 + w2*x2 + b. For (1,1): z = 1+1-1.5 = 0.5 >= 0 -> output 1. All other options give z < 0.

Q11. Perceptron misclassifies x=[2,3], t=1, ■=-1 ({-1,+1} encoding), η=0.5, w=[0,0]. Update rule: w += η(t-■)x. Find updated w.

Answer: c. w = [2, 3]

Explanation: ∆w = η(t-■)x = 0.5*(1-(-1))*[2,3] = 0.5*2*[2,3] = [2,3]. New w = [0,0]+[2,3] = [2,3].

Q12. Binary classifier: linear activation + squared error on 0/1 labels vs sigmoid+cross-entropy. Main practical issue with linear+squared-error?

Answer: b. Predictions can fall outside [0,1] and gradients no longer align with a probability-based decision boundary

Explanation: A linear output is unbounded, so it can predict values <0 or >1, which is meaningless as a probability, and the squared-error gradient doesn't correspond to a proper log-likelihood/probabilistic objective.

Q13. Linear regression: w=[1.0,2.0]■, ∇J=[0.4,0.8]■, η=0.5. Find w after one GD step (w_new = w - η∇J).

Answer: a. [0.8, 1.6]■

Explanation: w_new = [1.0,2.0] - 0.5*[0.4,0.8] = [1.0-0.2, 2.0-0.4] = [0.8, 1.6].

Q14. RMSE=4.5 lakhs, R^2=-0.08. Stakeholder says 'R^2 near zero means model is neutral.' Correct response?

Answer: d. $R^2 < 0$ means the model is worse than simply predicting the mean price for every house — it should not be deployed

Explanation: $R^2 = 1 - (SS_{res}/SS_{tot})$. A negative R^2 means the model's squared error exceeds that of a trivial mean-predictor baseline, i.e. it's actively worse, not 'neutral'.

Q15. Single-layer perceptron fails on XOR. Will adding more neurons to the same single layer fix it?

Answer: c. No — a single layer (any width) still produces a linear decision boundary; XOR needs a multi-layer architecture

Explanation: Neurons in a single layer without a following nonlinear combination layer only compute a linear combination of inputs; XOR is not linearly separable, so no amount of single-layer width alone solves it — you need a hidden layer + nonlinearity + combining output layer.

Q16. Perceptron, η=1, w=[0,0], b=0. Samples in order: (1,1)->1; (-1,1)->0; (1,-1)->0; (-1,-1)->1. Apply w+=η(y-■)x, b+=η(y-■) only on misclassification. Find final w, b.

Answer: c. w = [0,0], b = 0

Explanation: Step 1: (1,1),y=1: z=0, ■=0 (since not >0) -> misclassified; w=[0,0]+1*(1-0)*[1,1]=[1,1], b=0+1=1.
Step 2: (-1,1),y=0: z=-1+1+1=1>0 -> ■=1; misclassified; w=[1,1]+1*(0-1)*[-1,1]=[1+1,1-1]=[2,0], b=1-1=0.
Step 3: (1,-1),y=0: z=2*1+0*(-1)+0=2>0 -> ■=1; misclassified; w=[2,0]+1*(0-1)*[1,-1]=[2-1,0+1]=[1,1], b=0-1=-1.
Step 4: (-1,-1),y=1: z=1*(-1)+1*(-1)-1=-3, not>0 -> ■=0; misclassified; w=[1,1]+1*(1-0)*[-1,-1]=[1-1,1-1]=[0,0], b=-1+1=0.
Final: w=[0,0], b=0.

Q17. MLP: 2 hidden neurons with LINEAR (identity) activation feeding a single linear output, for XOR. Can correct weight tuning solve it?

Answer: d. No — a composition of linear maps is still linear, regardless of hidden width

Explanation: Stacking only linear layers (no nonlinearity) is mathematically equivalent to a single linear transformation, y = W2(W1 x + b1) + b2 = W'x + b'. Since XOR isn't linearly separable, this architecture can never solve it, no matter how it's tuned.

Q18. Predict house price from bedrooms, location, sqft using known past prices — best learning setup?

Answer: a. Supervised regression

Explanation: The target (price) is a continuous known label paired with input features → supervised regression.

Q19. h=ReLU(w1*x+b1), ■=w2*h+b2, L=½(■-y)². x=1.0, w1=2.0, b1=-1.0, w2=3.0, b2=0.5, y=2.0. Find ∂L/∂w1.

Answer: d. 4.5

Explanation: Forward: z1=w1*x+b1=2*1-1=1; h=ReLU(1)=1; ■=w2*h+b2=3*1+0.5=3.5.
∂L/∂■=(■-y)=3.5-2=1.5. ∂■/∂h=w2=3. ∂h/∂z1=1 (since z1>0, ReLU derivative=1). ∂z1/∂w1=x=1.
Chain rule: ∂L/∂w1 = 1.5 * 3 * 1 * 1 = 4.5.

Q20. Single-neuron regression ■=wx+b, x=2, y=5, w=1.5, b=0.5, α=0.01, L=(■-y)². ∂L/∂w=2(■-y)x, ∂L/∂b=2(■-y). Find updated w, b.

Answer: d. w = 1.56, b = 0.53

Explanation: ■ = 1.5*2+0.5 = 3.5. Error = ■-y = 3.5-5 = -1.5.
∂L/∂w = 2*(-1.5)*2 = -6 -> w_new = 1.5 - 0.01*(-6) = 1.5+0.06 = 1.56.
∂L/∂b = 2*(-1.5) = -3 -> b_new = 0.5 - 0.01*(-3) = 0.5+0.03 = 0.53.